

Delphes vs. CMS for $HH \rightarrow b\bar{b}\tau\tau$: where we stand, and one decision to take

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Abstract

The $\tau_h\tau_h$ yield deficit that motivated this work (a factor ~ 5) is closed: the cutflow agrees with CMS at every stage (1.01) and eight of ten NSBI features agree. The two that do not, $m_{\tau\tau}$ and $\Delta R_{\tau\tau}$, are both on the τ side. The $m_{\tau\tau}$ offset is now traced: it is in the τ four-vectors, fakes are excluded, and the Delphes τ energy response carries a one-sided tail CMS does not have — at gen $p_T = 20\text{--}30\text{ GeV}$, one Delphes τ in ten has more than twice the median response, against 1.08 on CMS. A multiplicative median map cannot reshape a distribution, so the current correction cannot close this. That leaves three options, priced in Section 2; the measurements are in Section 4.

1 Status

Stock CMS card plus the patches in `docs/card_patch_validation.md`. All tuning is *downstream*, so samples are never re-made; every map is measured on a private 2024 NanoAODv15 anchor, so “tuned” means “measured on CMS and transferred”.

quantity	status	how
b-tag efficiency (b, c, light)	✓ tuned	stochastic re-tag from <code>Jet.Flavor</code> at anchor UParT-AK4 Medium
b-jet energy scale	✓ tuned	inverted to unity against <code>GenJet</code> (self-anchored)
m_{bb} peak and width	✓ matched	follows
τ_h ID efficiency	✓ tuned	anchor <code>GenVisTau</code> → <code>DeepTau v2.5</code> Medium; includes HPS reco inefficiency
τ_h visible mass	✓ tuned	sampled per jet from anchor per- p_T quantiles
τ_h acceptance, jet- τ overlap	✓ identical	$p_T > 20$, $ \eta < 2.3$; pair first, then $\Delta R > 0.4$
p_T^{miss} resolution	✓ tuned	Gaussian smearing, 16.5 → 32.6 GeV/component
lepton reco efficiency	✓ tuned	per- p_T SFs ($1.57 \rightarrow 1.22$ e ; $1.30 \rightarrow 1.13$ μ)
di- τ mass estimator	✓ identical	same covariance-free FastMTT both sides
τ_h energy scale	✗ tuned	CMS/Delphes response ratio — under-corrects, Section 4.1

Table 1: Object status. Every map is derived on the anchor and applied downstream; none needs a card change.

✓ agree	m_{HH} (the κ_λ observable, across $\kappa_\lambda = 0, 1, 5$); m_{bb} (both peak $\sim 112\text{ GeV}$); p_T^{HH} ; $\Delta\phi_{HH}$; p_T^{H1}, p_T^{H2} ; ΔR_{bb} (a
✗ disagree	$m_{\tau\tau}$ ($\sim 7\text{ GeV}$ high, Delphes broader); $\Delta R_{\tau\tau}$ (excess at 2.2–3.2)

Table 2: NSBI inputs at $\kappa_\lambda = 0$; other κ_λ points behave the same. Yields: gen-matched cutflow 1.01, analysis-level $1.18\times$.

2 The decision

A CMS τ_h is an HPS object built from a narrow set of decay products; a Delphes τ_h is an $R = 0.4$ jet carrying **TauTag**. Mass, energy and direction are therefore wrong at source and we correct the first two rather than fix the representation. Section 4.1 shows the correction cannot succeed as built: a multiplicative per- p_T median cannot depopulate a kinematically forbidden region, and the Delphes response is one-sidedly broader at matched median by $3\times$ at low p_T . That is narrower than “only rebuilding works” — hence three options, not two.

The three options have now been costed and their accuracy quantified: throughput and storage by benchmark, the accuracy axis by a toy that propagates the *measured* response quantiles (Table 4) into $m_{\tau\tau}$ for $H(125)$, $Z(91)$ and a $t\bar{t}$ continuum. Scale context: the full campaign is $\sim 300\text{M}$ events ($3\times 5\text{M}$ signal, $3\times 50\text{M}$ $t\bar{t}$, $\sim 100\text{M}$ DY), and the fake- τ_h population inverts there — in the signal selection fakes are 4% and demonstrably irrelevant, but in **TTto4Q** every selected τ_h is a fake by construction, and fakes currently keep their raw jet p_T .

	(A) HPS rebuild	(B) response resampling	(C) accept
dev time	3–5 weeks	~ 1 day (+1–2 fake map)	0
extra CPU / I/O / storage at 300M	~ 0 / +13% / 0	+0.02% / 0 / 0	0
$m_{\tau\tau}$ fidelity vs CMS (real τ)	$\sim 96\%$ (assumed)	100% by construction	90% (84–92)
fake- τ_h energies	fixed by construction	fixed by the fake-map extension	broken, no reweight
τ direction	only option that touches it	unchanged	unchanged

Table 3: The three options at the 300M-event production scale. Fidelity = fraction of the CMS $m_{\tau\tau}$ S/B sensitivity reproduced (1D Fisher bound; see text).

(A) Rebuild the τ_h from particle flow. The samples retain the EFlow candidates (32 TB), so A needs no re-production; the cone kernel benchmarks $5\text{--}20\times$ above the pipeline wall and the added read volume is $\sim 13\%$, so A’s cost is purely 3–5 weeks of development and 10–25 tuning iterations (cone size, strip merging, π^0 threshold). Its risks are now itemised: 60–70% likelihood that tower-granular neutrals (0.017×0.017 , but with smeared centres) make the rebuilt *mass* no better than the sampled one, and 50–60% that a residual energy-scale map returns anyway (a narrow cone removes contamination, but Delphes calorimeter thresholds still differ from CMS PF). Its unique, B-inaccessible deliverable is the τ direction — the one lever on $\Delta R_{\tau\tau}$, which no energy correction reaches.

(B) Correct the distribution, not the scale. For each gen-matched τ -jet set $p_T^{\text{reco}} = p_T^{\text{gen}} \times r$ with r drawn from the CMS response quantiles — the machinery already used for the visible mass. By construction the gen-matched response then *is* the CMS response: 100% fidelity, exactly. The measured marginal cost is 0.02% of pipeline CPU. Three extensions close its gaps: a *fake-response map* (CMS fake response measured on the $t\bar{t}$ anchor already on disk: gen-unmatched **Tau** vs matched **GenJet**) recovers the fake-energy mismodeling in full; *vectorial MET propagation* (subtract the old τ vectors, add the new) closes the balance objection; and one *transfer check* of the CMS response quantiles between the HH and $t\bar{t}$ anchors (medians within 1%, quantile ratios within 5%) validates using one map across processes — if DY is doubted, a 200k-event DY anchor is a $\sim 2\text{--}3$ GB download. What B leaves untouched: the τ direction, decay mode, and charge.

(C) Accept and document — now excluded by measurement. With medians perfectly aligned, the residual *shape* alone widens H by 21% and Z by 33%, drops the H -vs- Z AUC from 0.976 to 0.905, and multiplies $Z \rightarrow \tau\tau$ leakage into the H window **by 5.2** (2.5% $\rightarrow$ 13.3%). A 1D Fisher fit puts C at 90% (84–92) of the CMS $m_{\tau\tau}$ sensitivity. The DY leakage would be visible in the paper’s own $m_{\tau\tau}$ figure — a reviewer problem independent of the κ_λ number.

Reading the fidelity numbers. They are 1D bounds on the $m_{\tau\tau}$ -mediated share of the κ_λ error, not the κ_λ error itself (κ_λ information lives mostly in m_{HH} , identical across options). Two further caveats: the response’s *lower* half is unmeasured (all scenarios share a mirrored CMS-like lower tail), and A’s 96% assumes a 30% core widening from tower granularity — a labelled guess. The fake near-cancellation in the 1D fit (0.5–1.7%) is toy-specific; a 10-feature NSBI sees the full +16 GeV / 20%-width fake-shape mismodeling (13% of the dominant background’s shape non-overlapping), which is why the fake map is not optional for B.

Cost is not the axis. The full 300M-event ntuplization is ~ 185 core-hours flat, and ~ 50 with the (verified bit-identical) restricted-ancestry walk; DY events are $\sim 10\times$ lighter still. Storage: the ntuplizer currently writes the full gen record — 99.5% of a 9 TB output; pruning to τ ancestry + bosons cuts it to ~ 0.4 TB. Both worth doing under any option. The real trade is *weeks of development and tuning judgement (A) versus a day (B)*, against the accuracy column above.

Recommendation. **B+ now** (resampling, fake map, MET propagation, anchor transfer check): $<1\%$ $m_{\tau\tau}$ fidelity error for the whole campaign, this month. **Not C. Gate A on the one thing only it can fix:** measure the τ *axis* residual (jet axis vs gen-visible- τ axis, both tiers — the same probe pattern as the response study, angle instead of energy; a few hours on existing files). If Delphes’ axis residual is large where CMS’s is small, A has a unique quantified justification for $\Delta R_{\tau\tau}$ and the weeks become defensible; if not, B+ closes the case. Keep the 32 TB of EFlow until that measurement is done.

Questions for τ expertise.

1. Is tower-granular neutral reconstruction adequate for a useful decay-mode assignment, or does it degrade π^0 content enough that the rebuilt mass is no better than the anchor-sampled one?
2. Signal cone: fixed 0.1 or p_T -dependent, over the 20–60 GeV range that dominates here?
3. Seeding: jets (as HPS) or direct PF clustering? Would either change the efficiency for collimated pairs, $\Delta R_{\tau\tau} \lesssim 1$?
4. Is the $\Delta R_{\tau\tau}$ pattern — over-efficient above 2.5, possibly under-efficient below 1 — recognisable as a jet-versus-HPS effect?
5. Is option (B) acceptable in a pheno paper, or does resampling a reconstructed quantity undercut the claim?
6. Is a $1.18\times$ analysis-level yield excess acceptable for unbinned NSBI, where training uses shapes and normalisation is fitted?

3 Known limitations

In four buckets, because “known” was covering different things.

Accepted by design. FastMTT is covariance-free (a fixed p_T^{miss} resolution rather than the per-event covariance), but is run identically on both sides, so it cancels in the comparison and bounds the absolute $m_{\tau\tau}$ resolution rather than the agreement. The card runs without pileup, so jet energy *resolution* is untuned and pileup jets cannot exist; both act on the $b\bar{b}$ side, which agrees, and the second cannot be fixed downstream in any case. p_T^{miss} smearing is isotropic and flat where the real resolution is anisotropic and grows with ΣE_T : this reproduces the width but not its structure, and it remains a candidate for $m_{\tau\tau}$ being marginally *broad* on Delphes — Section 4.1 exonerates p_T^{miss} for the *offset*, not for the width.

A scope bound, not a defect. **taumistag** is measured on the b -rich signal anchor, and fake rates are strongly jet-flavour dependent, so it is *not* valid for the fake-dominated backgrounds

(QCD, W+jets). This does not limit the present signal study, but the samples cannot support a background-heavy analysis until those are measured separately.

Real defects, not yet fixed. Two in the energy-scale *application*: fake τ_h match neither branch of `escale_factor` and are left uncorrected while still receiving a τ -like mass, and the τ branch matches *all* gen τ (`|pid|==15`) including leptonically decaying ones, whereas the map is derived on the hadronic visible set. Both were measured *not* to be the cause of the $m_{\tau\tau}$ offset (Section 4.1), but both are wrong.

Untested — risks, not accepted limitations. **(a)** The sampled τ_h mass is drawn independently of the τ energy and of the partner leg, whereas the true visible mass correlates with decay mode and hence with prong multiplicity and energy sharing. Size unknown. **(b)** Every map is flat-extrapolated above 300 GeV, the last bin edge, so a 500 GeV jet is given the 200–300 bin’s value verbatim. Four reasons this is probably benign, none of them yet checked directly:

- *Population.* Only $\sim 2.8\%$ of b-jets exceed 300 GeV.
- *It is not where κ_λ lives.* The $gg \rightarrow HH$ amplitude is $\kappa_\lambda \times (\text{triangle}) + (\text{box})$, and the triangle carries an s -channel Higgs propagator falling with $\hat{s}$. The m_{HH} *shape* difference between κ_λ hypotheses is therefore largest near threshold, $m_{HH} \simeq 250\text{--}400$ GeV, where the destructive interference is most modified; the high- m_{HH} tail is box-dominated and comparatively κ_λ -insensitive. b-jets from the discriminating region sit well below 300 GeV. (An earlier version of this note asserted the opposite and used it to argue the extrapolation was dangerous; that was wrong.)
- *Flat is the right asymptote.* The v1 card’s b-tag formula, $(p_T > 4) \times (0.904 - 3.53/p_T)$, tends to 0.904, so a constant extrapolation matches the underlying behaviour rather than being arbitrary.
- *The escale has little left to do there.* Table 4 shows the Delphes response converging on CMS by 100–300 GeV (1.035 vs 0.970), so the correction is small and slowly varying where the extrapolation begins.

What would change this is a boosted or high- m_{HH} category, where the extrapolated region stops being a 3% tail and becomes the signal region.

4 The two open questions

4.1 $m_{\tau\tau}$

$m_{\tau\tau} = m_{\text{vis}} / \sqrt{x_1 x_2}$ factorises, so the *visible* di- τ mass m_{vis} — the two τ four-vectors, no p_T^{miss} , no FastMTT — separates a τ scale error from an x -fit error. The two have disjoint fixes.

It is the four-vectors. m_{vis} is offset by *at least* as much as $m_{\tau\tau}$ (peak ratios ≈ 1.24 vs ≈ 1.17), so FastMTT is not the source; if anything it compresses. Both legs are individually harder. With an applied scale of $\simeq 0.85$ and a residual $\simeq 1.24$, the raw Delphes τ are $\sim 46\%$ harder than CMS while the map corrects 15%.

Not fakes. Splitting gen-matched from fake gives fake fractions 4.1% Delphes vs 4.3% CMS with indistinguishable shapes; the gen-matched component carries everything — the peak displacement and an unphysical tail at $m_{\text{vis}} > 125$ GeV, kinematically impossible for $H \rightarrow \tau\tau$ and absent on CMS. Same pattern as $\Delta R_{\tau\tau}$.

The response distribution. Plotting the response (reco τ p_T / gen-visible- τ p_T) as a *distribution*, each side divided by its own median so the scale is removed by construction, leaves only what the map cannot change. One control first: without a τ ID, Delphes matches a gen τ to the nearest jet of *any* kind (so a lost τ -jet can match a b -jet), while CMS matches within the Tau collection — and mis-matching mimics the same p_T dependence. With the ID required on both sides the tail survives (Table 4, Figure 1):

CMS is flat in p_T (1.10–1.14), as a calibrated object with constant fractional resolution should be ($q_{90}/\text{median} = 1.076 \Rightarrow \sigma \simeq 6\%$, right for HPS), so the anchor is sound. Delphes is not flat and

gen p_T [GeV]	median		q_{95}/median		ratio
	Delphes	CMS	Delphes	CMS	
20–30	1.184	0.989	3.378	1.135	$3.0\times$
30–50	1.114	0.987	1.873	1.105	$1.7\times$
50–100	1.066	0.978	1.325	1.102	$1.2\times$
100–300	1.035	0.970	1.195	1.115	$1.07\times$

Table 4: τ_h energy response, τ ID required both sides. q_{95}/median is pure *shape*: the scale is divided out.

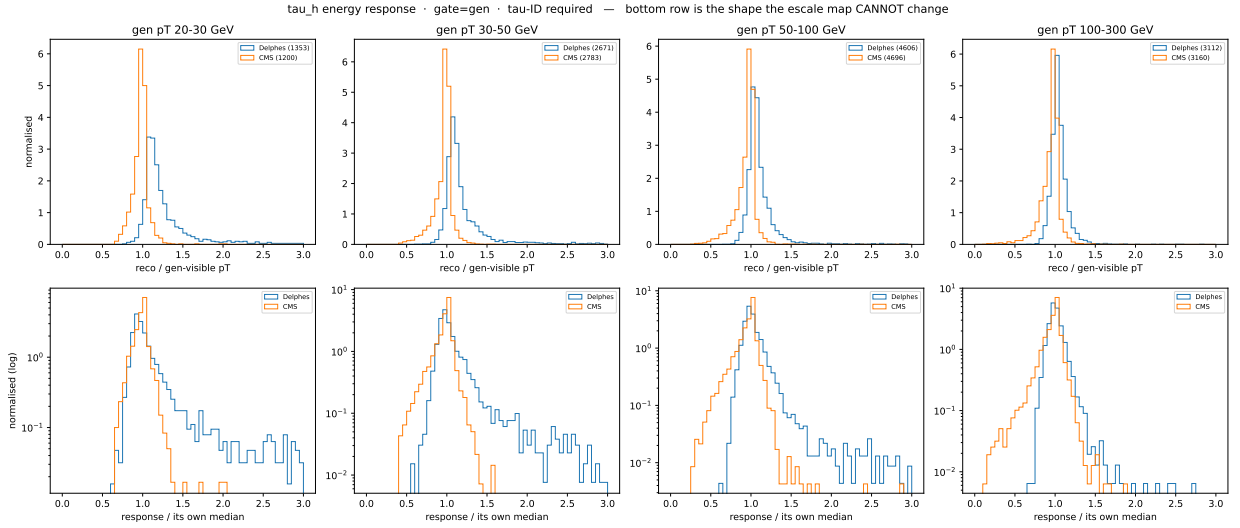


Figure 1: τ_h energy response per gen- p_T bin, τ ID required on both sides. **Top:** as measured — the offset `tau_scale` is built to remove. **Bottom (log):** each side divided by its *own* median, so the scale is gone by construction and only the shape remains — i.e. only what a multiplicative map *cannot* change. The Delphes shelf above $1.5\times$ shrinks with p_T and is consistent with CMS by 100–300 GeV; CMS is flat throughout. Numbers in Table 4.

converges on CMS as p_T rises — what a constant *absolute* energy excess gives, large as a fraction at 25 GeV and small at 175. Medians converge alongside, as one cause predicts. Sharpest number: at 20–30 GeV the Delphes q_{90}/median is 2.156 against 1.076 on CMS. That is not a resolution. One caveat survives: **TauTag** is a stochastic draw uncorrelated with contamination while DeepTau actively rejects it, so part of the gap is a *selection* correlation — equally unreachable by a scale map, so it does not change Section 2, but the tail is not purely about cone size.

Excluded. (i) *The sampled mass.* The FastMTT floor $x_{\min} = (m_{\text{vis}}^{\text{leg}}/m_\tau)^2$ exceeds a typical fitted $x \simeq 0.5\text{--}0.9$ only above ~ 1.26 GeV, so below that the mass is inert: displacing the whole low-mass region by 200 MeV moves $\langle m_{\tau\tau} \rangle$ by 0.00%, against -2.4% in the a_1 tail. Replacing the 21-quantile grid with an atom at $m_{\pi^\pm}$ plus a histogram CDF changes it by 0.11%. (ii) *A symmetric resolution mismatch.* $7\% \rightarrow 25\%$ at fixed median moves the mean 0.05% (RMS $+2\%$) — but that test used a symmetric log-normal, so it does *not* cover the one-sided tail above.

4.2 $\Delta R_{\tau\tau}$: a selection effect, cause unknown

Gen-level $\Delta R_{\tau\tau}$ spectra agree, so the samples are the same physics; the efficiencies differ — Delphes selects 0.22–0.23 at gen $\Delta R > 2.5$ where CMS selects 0.15–0.17 ($\sim 3\sigma$), with a weaker ($\sim 2\sigma$) hint of loss below $\Delta R < 1$. Both push the distribution rightward. Excluded by measurement: fakes (the discrepancy is entirely gen-matched); τ -pair selection (symmetrising moved $\Delta R_{\tau\tau}$, not $m_{\tau\tau}$); CMS τ miscalibration (anchor response 0.97–0.99). This matters more than its size: it is a *selection* effect, so no downstream correction reaches it, and unlike $m_{\tau\tau}$ we have no candidate mechanism.

A Reference

All reproducible from `delphes-pipeline`.

1. **Cutflow** (`tauh.cutflow.py`). Five stages — gen visible τ_h , reco, ID, cleaned jets, FastMTT — walked identically both sides on *conditional* efficiencies. FastMTT once carried the whole deficit (12% vs 99%); now 1.01 overall. Tests inject a loss at one stage and assert it is attributed there only.
2. **FastMTT, and why the mass is sampled.** The hadronic prior vanishes unless $x_{\min} \leq 1$; an AK4 jet mass (5–20 GeV vs $m_\tau = 1.777$) admits no solution, returning NaN and silently removing $\sim 89\%$ of pairs. The anchor mass fixes it, but must be *sampled*: it is a one-sided floor, so a single median relaxes a real constraint for every leg above it, inflating the width $\sim 60\%$ and mean $\sim 9\%$ while leaving the *mode* unchanged — a mode-gated $Z \rightarrow \tau\tau$ candle would miss it.
3. **The reference error, twice.** A GenJet is also a jet and also contains UE, so it is not the analogue of GenVisTau. Against a true visible τ ($p_\tau - p_{\nu_\tau}$, exact for one neutrino) the Delphes τ -jet is 17% harder at 25 GeV, $\sim 2\%$ at 250. The same error appeared in the escale map and in the cutflow’s stage 1.
4. **Order of application.** Maps measured at CMS scale must be applied at CMS scale; tagging raw jets evaluated a steep efficiency $\sim 17\%$ too far right. Applying the scale first moved the excess $1.25 \rightarrow 1.18$.
5. **Gen- τ check** (`gen_tau_check.py`). Our visible- τ construction vs CMS GenVisTau on the same events: efficiency 0.991, purity 0.998, stage-1 rate to $\sim 1.6\%$.
6. **m_{vis} and the split** (`nsbi_overlay.py --diagnostics --split-gen-matched`). m_{vis} uses neither p_T^{miss} nor the fit, so it attributes an offset to one of two disjoint causes; tests pin $m_{\text{vis}} \leq m_{\tau\tau}$, invariance under $p_T^{\text{miss}} 40 \rightarrow 200$ GeV, and exact linearity in the τ scale. The split normalises to each side’s total, so the fake curve’s area is the fake fraction.
7. **Response shape** (`tau_response.py`). Distribution per gen- p_T bin, lower row divided by each side’s own median. Tests assert the separation: a pure scale difference vanishes under

normalisation, an injected one-sided tail survives. `--gate` and `--require-id` expose two asymmetries in the shipped derivation (anchor gates on gen and conditions on DeepTau; Delphes gates on reco and conditions on nothing).

8. **Map accuracy.** No map is statistics-limited at 2×10^5 anchor events (b -tag 0.25–0.42%/bin, τ_h eff 0.43–1.26%, escale 0.2–0.3%, p_T^{miss} 0.16–0.21%); what limits a map is np , not n . Two defects do not improve with statistics: the mass quantile grid interpolates across the empty region between the $\pi^\pm$ atom and the ρ shoulder ($\sim 6\times$ over-populated, measured inert), and the 300 GeV flat extrapolation (untested).
9. **Option tradeoff analysis.** Costs by benchmark (resampling marginal cost, HPS cone kernel, restricted-ancestry walk — the last verified bit-identical against the current `gen_taus` on records with copy chains); accuracy by a common-random-number toy propagating the measured response quantiles into $m_{\tau\tau}$ for $H/Z/t\bar{t}$ (AUC, overlap, 1D Fisher), every headline number re-derived by an independent check. Result in Section 2.
10. **Generator copies.** Delphes writes the full Pythia history, CMS GenPart is pruned. Rule: anything that **counts** generator objects must collapse copies first; anything that **matches** geometrically need not.

Reproducing.

```
# --max-events is REQUIRED: derive_maps caps the anchor from the config but reads
# the Delphes side uncapped.
python -m delphes_pipeline.tuning.derive_maps --config config.v1.yml \
    --output cards/tuning/maps_v1.json --max-events 200000
python -m delphes_pipeline.validation.run_validation --config config.v1.yml
python scripts/tauh_cutflow.py --config config.v1.yml [...] --gen-dr plots/gen_dr
python scripts/nsbi_overlay.py --config config.v1.yml [...] --tautau-only \
    --diagnostics --split-gen-matched
python scripts/tau_response.py --config config.v1.yml [...] --require-id
```